15. Performance Testing

Date	1 July 2026
Team ID	SWTID-2026-4570
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance testing was carried out locally against the running FastAPI application using its TestClient, issuing 15 sequential requests per endpoint and measuring wall-clock response time. Because no GEMINI_API_KEY was configured for this test run, all requests were served from EduGenie's built-in demo/mock mode — this isolates and measures the overhead of the FastAPI routing, Pydantic validation and JSON-file logging layers, independent of live Gemini network latency.

Step 1: Testing Overview

Testing Tool Used	Python (time.perf_counter) via FastAPI TestClient, 15 requests per endpoint
Type of Testing	Baseline response-time measurement (demo/mock mode)
Target Module / API	/qa, /api/explain, /api/quiz, /api/summary, /learn/recommendations, / (home page)
Test Environment	Local container, single process, no live Gemini network calls
Test Date	10 July 2026

Step 2 & 3: Test Scenarios and Results (15 requests each, demo/mock mode)

Endpoint	Avg Response Time	Min	Max	Status
GET /qa	5.02 ms	3.79 ms	16.60 ms	Pass
POST /api/explain	4.36 ms	4.03 ms	5.16 ms	Pass
POST /api/quiz	8.90 ms	7.51 ms	13.81 ms	Pass
POST /api/summary	5.68 ms	5.14 ms	7.13 ms	Pass
GET /learn/recommendations	5.72 ms	5.35 ms	6.37 ms	Pass
GET / (home page)	4.76 ms	1.32 ms	41.61 ms	Pass (one outlier on first request — template compilation)

Step 4: Observations & Analysis

- All endpoints responded well within the 2-second target under demo/mock mode; the entire measured overhead is FastAPI routing + Pydantic validation + JSON-file logging, not AI inference.
- /api/quiz was the slowest of the AI endpoints (8.90 ms avg) because it also logs one record per generated quiz question to quizzes.json.
- The single 41.61 ms outlier on the home page was the first request, consistent with one-time Jinja2 template compilation.
- Live-mode performance (with a real GEMINI_API_KEY) was not measured in this test run and will be dominated by Gemini API network latency plus the built-in retry/backoff logic (up to 3 attempts); this is recommended as a follow-up test using a tool such as Locust or k6 once a production API key and hosting environment are available.
- The JSON flat-file data store performed adequately for this single-user, sequential test; concurrent-write behavior under multiple simultaneous users has not yet been load-tested and is flagged as a scalability risk (see Section 20, Scalability & Future Plan).